

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456559.70714 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.0052
Transit depth (Rp/Rs)^2	3.24 +/- 0.19 [%]
Orbital Inclination (inc)	89.58 +/- 0.99
Airmass coefficient 1 (a1)	1.205 +/- 0.011
Airmass coefficient 2 (a2)	-0.1495 +/- 0.0019
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	None
Optimal Aperture	3.86
Optimal Annulus	12.66
Transit Duration (day)	0.13393 +/- 0.0008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.0052 vs 0.1489 (deviation 5.36σ)
Duration fit vs published	0.13393 d vs 0.1304 d (deviation 3.94σ)
Mid-time O–C	2.8 min
Depth SNR	31.0
Residual scatter	0.95 %

Light curve

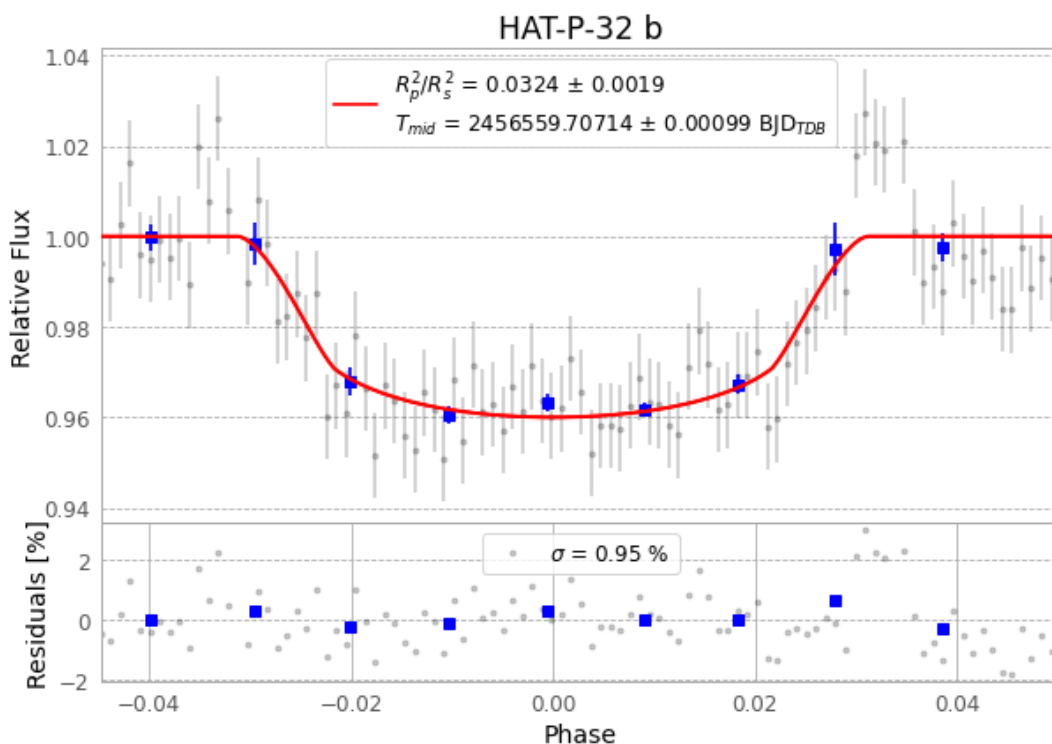


Figure 1 — FinalLightCurve_HAT-P-32 b_2013-09-23.png (SHA-256 cc88828fdcc8a074...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [207, 123], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a83fe3de03ad0f6d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

99 FITS frames (65 MB), HATP-32130924023313.FITS → HATP-32130924072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-130924.log (29c452df1816...), FinalLightCurve_HAT-P-32 b_2013-09-23.png (cc88828fdcc8...), FinalLightCurve_HAT-P-32 b_2013-09-23.csv (d6db82e9381a...)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 11:05Z.